

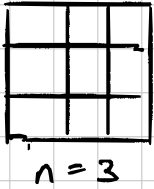
Assignment 4

Problem 3.6

Given: $n \times n$ warehouse grid

Part A: Propositional Encoding Cost

- 1) If $n=3$ as an example, the adjacency sentences will be:



$(1,1), (1,2), (2,1)$
 $(2,1), (2,2), (3,1), (1,1)$
 $(3,1), (2,1), (3,2)$
 $(1,2), (1,1), (2,2), (1,3)$
 $(2,2), (2,1), (1,2), (2,3), (3,2)$
 $(3,2), (3,1), (2,2), (3,3)$
 $(1,3), (1,2), (2,3)$
 $(2,3), (1,3), (2,2), (3,3)$
 $(3,3), (3,2), (2,3)$

See
Corrections
Below

There are 9 adjacency sentences

$(n^2)^2 \times \Rightarrow n^2$ number of adjacency sentences,

$\therefore$ as n grows, the number of adjacency sentences grows exponentially.

At $n=100$, number of adjacency sentences are $100^2 = 10000$

- 2) $D_{2,1} \Rightarrow C_{1,1}, C_{2,2}, C_{3,1}$
 $D_{3,1} \Rightarrow C_{2,2}, C_{3,1}$
 $D_{1,2} \Rightarrow C_{1,1}, C_{2,2}, C_{1,3}$
 $D_{2,2} \Rightarrow C_{2,1}, C_{1,2}, C_{2,3}, C_{3,2}$
 $D_{3,2} \Rightarrow C_{3,1}, C_{2,2}, C_{3,3}$
 $D_{1,3} \Rightarrow C_{1,2}, C_{2,3}$
 $D_{2,3} \Rightarrow C_{1,3}, C_{2,2}, C_{3,3}$
 $D_{3,3} \Rightarrow C_{3,2}, C_{2,3}$

There are 8 damage sentences because $(1,1)$ is safe.

$\Rightarrow n^2 - 1$ number of damage sentences

At $n=100$, number of damage sentences based on the creating rule $= 100^2 - 1 = 9999$

Corrections:

- 1) Adjacency: In propositional logic there is no binary predicate, so adjacency is implicit. If we wanted to state it explicitly, e.g. using propositions $A_{(i,j),(i',j')}$

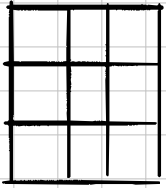
then we need an assertion for each ordered pair.

$$(n^2)^2 = n^8$$

So for $100 \times 100 : 10^8$ sentences

$$= n^2$$

3) Safety Rule.



Since (1,1) is always safe,
then number of safety sentences
needed using the safety rule is

$$n^2 - 1$$

$$\cancel{n^2}$$

Part B : FOL Comparison

- 1) Adjacency : Each square has at least one adjacent

✓ For every square L , there exists at least one adjacent square L' :

$$\forall L \exists L' \text{ Adjacent}(L, L')$$

Only one sentence needed regardless of n .

- 2) $\forall L \text{ Creaking}(L) \Leftrightarrow \exists L' \text{ Adjacent}(L, L') \wedge \text{ Damaged}(L')$

✓ For every location, L , that creaks, will only happen iff
there exists an adjacent location to L , L' ,
and that L' is damaged.

- 3) $\forall L \text{ Safe}(L) \Leftrightarrow \neg \text{ Damaged}(L) \wedge \neg \text{ Forklift}(L)$

✓

$$\text{or}$$
$$\forall L \text{ Safe}(L) \Leftrightarrow \neg (\text{ Damaged}(L) \vee \text{ Forklift}(L))$$

Part C : Translation Challenge

"There is exactly one damaged floor section in the entire warehouse"

$$\exists! L \text{ Damaged}(L) \wedge \forall L' \text{ Damaged}(L') \Rightarrow L' = L$$

There exists only one damaged floor and for every detected damage floor, L' , then that must be L .

Part C : Translation Challenge

FOL: The statement 'there is exactly one damaged floor section' is unice:

$$\exists! L \text{ Damaged}(L) \wedge \forall L' \text{ Damaged}(L') \Rightarrow L' = L$$

Propositional logic for 3×3 : With 9 damaged variables $D_{1,1}, D_{2,1}, \dots, D_{3,3}$ we need:

- At least one : $D_{1,1} \vee D_{2,1} \vee D_{3,1} \vee D_{1,2} \vee D_{2,2} \vee D_{3,2} \vee D_{1,3} \vee D_{2,3} \vee D_{3,3}$ — 1 clause
- At most one : For every pair of distinct variables, assert that they are not both true

$$\neg D_{i,j} \vee \neg D_{i',j'}$$

There are $\binom{9}{2} = 36$ such clauses.

Total : $1 + 36 = 37$ clauses for 3×3

Scaling for an $n \times n$ grid with n^2 damage variables, the at-most-one constraint requires $\binom{n^2}{2} = \frac{n^2(n^2-1)}{2}$ pairwise exclusion clauses

which is $O(n^4)$. For $n=100$: $\binom{10000}{2} = 49,995,000$ clauses. The FOL version remains 1 sentence regardless of n .

Part D: Practical Implications

1. Propositional at scale:

The basic physics encoding ($3n^2 = 30,000$ biconditionals for $n = 100$) is well within the capacity of modern SAT solvers, which routinely handle millions of clauses. However, richer constraints like "exactly one damaged floor" add $O(n^4) \approx 50$ million clauses which is feasible but pushing the limits. Adding multiple such cardinality constraints rapidly becomes unwieldy.

2. FOL restrictions for tractability

Unrestricted FOL is semi-decidable (and undecidable for satisfiability in general).

Tractable restrictions include

- (a) Horn clauses, which enable polynomial-time forward & backward chaining
- (b) Function-free FOL over finite domains, which is decidable,
- (c) Description Logics, which restrict quantifier patterns to guarantee decidability while retaining useful expressiveness
- (d) Bounded-variable fragments, which limit the number of variables.

3. Hybrid Approach:

Use FOL for the general physics rules
(3 quantified sentences, grid-size independent)

and propositionalize only in the local neighborhood around the agent's current position for fast inference.

For example, the agent could maintain a FOL KB for global constraints but ground the creaking/rumbling rules only for the squares within k steps of its current location. This combines FOL's compact representation with propositional logic's efficient (and decidable) inference. Z3 effectively does this internally via

MBQI (Model-Based Quantifiable Instantiation), grounding quantifiers on-demand.